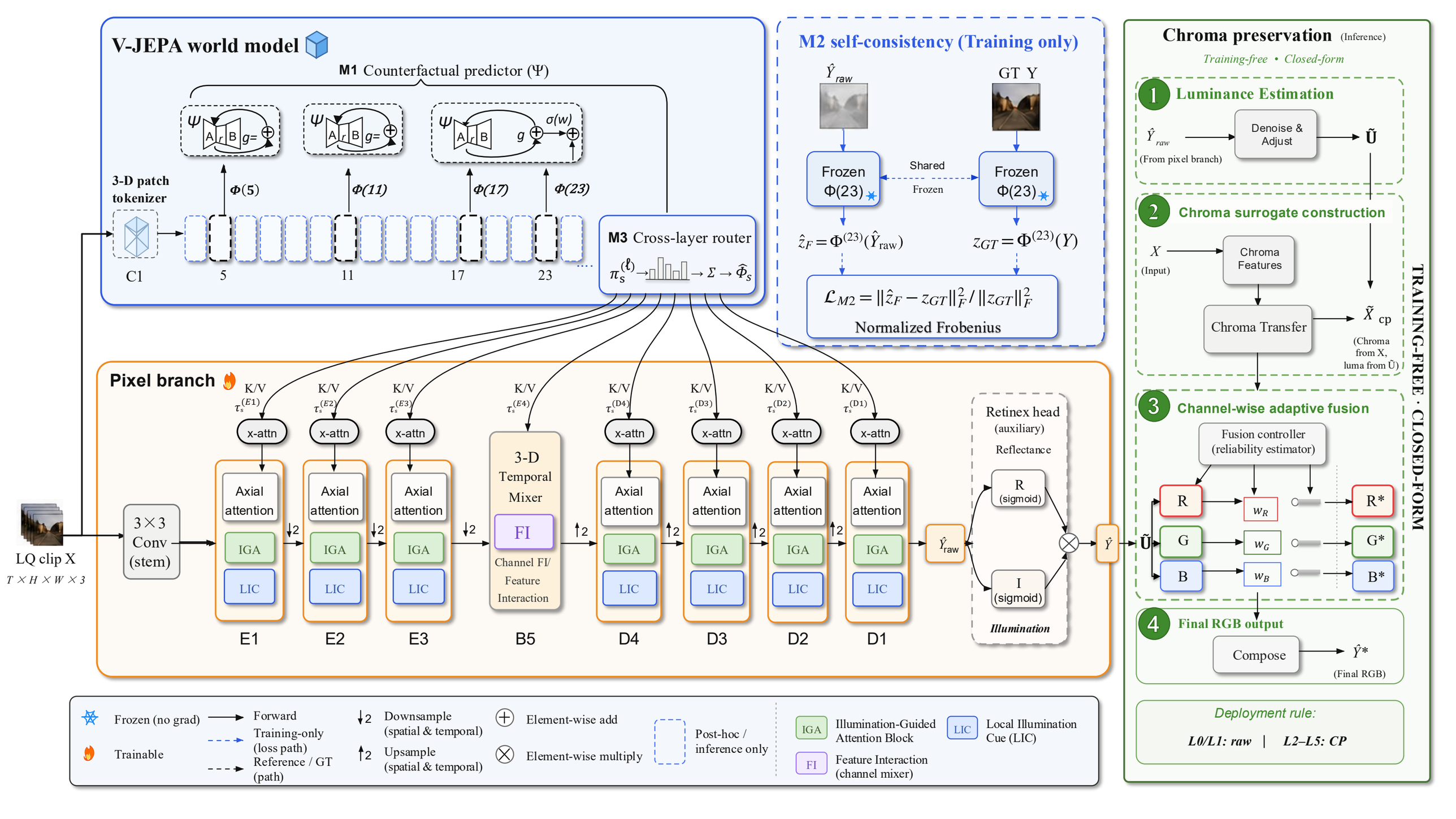

V-JEPA world model
M1 Counterfactual predictor (Ψ)
3-D patch tokenizer
C1
5
11
17
23
Φ(5)
Φ(11)
Φ(17)
Φ(23)
M3 Cross-layer router
M2 self-consistency (Training only)
Frozen Φ(23)
Shared
Frozen
GT Y
Normalized Frobenius
Chroma preservation (Inference)
Training-free • Closed-form
1 Luminance Estimation
Denoise & Adjust
(From pixel branch)
2 Chroma surrogate construction
(Input)
Chroma Features
Chroma Transfer
(Chroma from X, luma from Ũ)
3 Channel-wise adaptive fusion
Fusion controller (reliability estimator)
4 Final RGB output
Compose
(Final RGB)
Deployment rule:
L0/L1: raw | L2–L5: CP
TRAINING-FREE · CLOSED-FORM
Pixel branch
K/V
x-attn
Axial attention
IGA
LIC
3-D Temporal Mixer
FI
Channel FI/ Feature Interaction
Retinex head (auxiliary)
Reflectance
R (sigmoid)
I (sigmoid)
Illumination
E1
E2
E3
B5
D4
D3
D2
D1
LQ clip X
T × H × W × 3
3×3 Conv (stem)
Frozen (no grad)
Trainable
Forward
Training-only (loss path)
Reference / GT (path)
Downsample (spatial & temporal)
Upsample (spatial & temporal)
Element-wise add
Element-wise multiply
Post-hoc / inference only
Illumination-Guided Attention Block
Local Illumination Cue (LIC)
Feature Interaction (channel mixer)